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The dynamics of customer participation in service recovery: The roles of failure severity, quality signals, and responsiveness

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ABSTRACT

In high-contact hospitality settings, service failures can disrupt encounters in ways that prompt customers to intervene. Drawing on control theory and signaling theory, we conceptualize customer participation (CP) as a solution-oriented behavioral response through which customers attempt to influence recovery when service delivery is disrupted. We examine how quality signals and frontline employee responsiveness shape the extent to which participation translates into recovery outcomes. Across five studies, we show that severe service failures can prompt solution-oriented participation, but this response is contingent on contextual conditions. Quality signals attenuate the likelihood that failure severity translates into participation by shifting perceived responsibility toward the firm, creating a conditional quality signaling paradox. Employee responsiveness strengthens the positive effects of participation once enacted, but cannot compensate when participation is not initiated. These findings advance an understanding of CP by distinguishing between participation initiation and effectiveness, with managerial implications for balancing quality signaling and responsive frontline behavior.

1. Introduction

Imagine a couple celebrating their anniversary at a prestigious restaurant chosen for its awards and strong reputation. Upon arrival, they discover that their reservation is missing and no table is available. Trusting the venue's reputation, they initially wait in silence, expecting the issue to be resolved. As time passes with vague reassurances and no visible progress, frustration grows. Eventually, the couple speaks up, explaining the importance of the occasion, sharing preferences, and suggesting alternatives. Their intervention reflects not an intent to collaborate, but a recognition that recovery requires their input to progress.

Service failures are inevitable in hospitality contexts, where interactions are high-contact, emotionally charged, and often tied to milestone events such as celebrations or long-anticipated experiences (Bitner et al., 1990; Zeithaml et al., 1993). While service recovery research has traditionally focused on firm-led actions to restore satisfaction, it has paid comparatively less attention to customer-initiated participation during recovery as a mechanism shaping outcomes. Yet,

customer participation during recovery is not uniform; customers do not always intervene to articulate their needs, offer suggestions, or propose solutions.

These behaviors fall under the broad construct of customer participation (CP), which has typically been framed as proactive, firm-invited value co-creation (Bendapudi and Leone, 2003; Dong et al., 2008). This dominant framing, however, does not fully capture the situational nature of CP in failure settings. When customers intervene under conditions of severe disruption, their actions are often not motivated by engagement or shared responsibility, but by the recognition that progress depends on their involvement. Participation in these situations is prompted not by an invitation to co-create, but by the need to engage with a stalled recovery process.

This research extends service recovery theory by explaining a decision point that dominant frameworks have largely overlooked: when customers choose to intervene during recovery, rather than how they evaluate outcomes after recovery has occurred. Expectancy-disconfirmation (Oliver, 1980) and justice-based accounts (e.g., Tax et al., 1998) primarily explain post-hoc judgments of satisfaction and

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fairness. In contrast, we conceptualize CP during recovery as a solution-oriented behavioral response that emerges while the service encounter is still unfolding. Our framework identifies the conditions under which customers are more or less likely to engage in CP during recovery. By focusing on participation as a situationally triggered behavioral response rather than an evaluative reaction or a firm-invited role, we offer a process-oriented account of customer action during service recovery.

Drawing on control theory (Carver and Scheier, 1981; Hui and Bateson, 1991), we propose that CP arises when customers experience a disruption to the service encounter that creates a gap between their desired state and the unfolding experience, threatening their sense of control in pursuing their goals. This perceived loss of control heightens customers' motivation to intervene. In this context, CP functions as a solution-oriented behavioral response through which customers seek to reassert control by articulating needs, offering suggestions, and actively engaging with employees to shape the resolution. Such action is aimed at influencing the recovery process and preventing further deterioration, rather than co-creating value.

Importantly, reactive participation is not determined solely by the occurrence of failure itself. It is shaped by contextual cues that signal whether customer involvement is expected, necessary, or appropriate. We focus on two such cues: quality signals and employee responsiveness. Drawing on signaling theory (Spence, 1973), we argue that quality signals such as awards, certifications, or platform recognitions may attenuate the likelihood that failure severity translates into CP by shifting perceived responsibility for recovery toward the firm. In contrast, employee responsiveness shapes the extent to which participation translates into improved recovery outcomes once it occurs, by signaling acknowledgment and action on customer input. Across five studies, this research contributes to the service recovery and hospitality literatures by offering a process-oriented account of CP, clarifying when it emerges, the conditions under which it occurs, and how contextual cues shape both participation initiation and its effectiveness.

2. Proposed framework and hypotheses development

2.1. Customer participation as a solution-oriented response in service recovery

Customer participation (CP) in service recovery has traditionally been conceptualized as a co-creative process in which firms and customers jointly solve problems to enhance satisfaction and value (e.g., Dong et al., 2008; see Supplement file A for a review). This perspective emphasizes proactive, firm-invited involvement and assumes that customers participate when they are willing and prepared to do so. However, emerging research suggests that CP can also arise as a reactive response, driven not by a desire to co-create but by situational breakdowns that compel customers to intervene when service delivery falters (Joosten et al., 2017; Guo et al., 2016).

Building on this insight, we define CP as a solution-oriented behavioral response through which customers contribute input aimed at influencing how recovery unfolds. This includes behaviors such as suggesting alternatives, articulating preferences, and engaging in problem-solving. This framing distinguishes CP from two adjacent constructs. First, unlike proactive, firm-facilitated co-creation (Vargo and Lusch, 2016; Chan et al., 2010), CP in recovery contexts is situationally triggered rather than role-invited. Second, unlike complaint or voice behaviors, which primarily involve communicating dissatisfaction or assigning blame (Bell and Luddington, 2006), CP reflects efforts directed toward influencing how the problem is addressed. This distinction is particularly salient in hospitality contexts, where service delivery unfolds in real time and customers often lack predefined pathways for redress (Kim and So, 2023).

Control theory provides a useful lens for explaining when such participation becomes more likely. Service failures create a discrepancy

between customers' desired state and the unfolding experience, particularly when recovery progress appears uncertain or stalled (Carver and Scheier, 1981; Hui and Bateson, 1991). This discrepancy motivates corrective action aimed at restoring alignment with one's goals. Importantly, such disruptions can give rise to multiple responses apart from CP, including withdrawal, exit, or complaint (Khamitov et al., 2020). In contrast, we focus on CP as a solution-oriented behavioral response through which customers attempt to engage with a disrupted recovery process.

As failure severity increases, the perceived stakes of the situation become more salient, heightening customers' motivation to intervene to prevent further loss or misalignment (Roggeveen et al., 2012). Under these conditions, customers may be more likely to engage in CP by articulating needs or offering suggestions. We therefore expect that greater failure severity increases the likelihood that customers engage in CP, while recognizing that this response remains contingent on contextual conditions examined in subsequent sections.

Hypothesis 1. Service failure severity increases the likelihood that customers engage in customer participation during service recovery.

2.2. Customer participation as a process mechanism in service recovery

Service recovery research has demonstrated that customer participation (CP) can influence outcomes such as satisfaction and word-of-mouth when customers contribute to problem solving (e.g., Dong et al., 2008; Hazée et al., 2017). Prior work, however, has predominantly examined CP as a direct predictor of recovery outcomes, leaving its role within the recovery process itself less clearly specified. We extend this perspective by positioning CP as a solution-oriented behavioral response that operates as a process-relevant pathway linking service failure severity to recovery outcomes. This repositioning does not preclude CP from functioning as a direct predictor; rather, it specifies how failure severity translates into recovery outcomes through customer action.

Drawing on control theory, we locate CP within a broader self-regulatory process rather than treat it as a standalone driver of positive outcomes. Service failures, particularly those of high severity, generate a discrepancy between customers' desired and actual states, thereby activating a motivation to restore control (Carver and Scheier, 1981). This motivation arises from the discrepancy itself rather than from an expectation that any particular response will prove effective. Accordingly, CP is not premised on a belief that participation will necessarily succeed; rather, it reflects an attempt to engage with a disrupted recovery process under conditions of threatened control.

Within this process, CP represents the action through which customers respond to the discrepancy by articulating preferences, offering suggestions, or engaging with employees regarding the resolution. These behaviors constitute a pathway through which customer input enters the unfolding recovery process. Importantly, this pathway does not offset the direct negative impact of service failure on customer evaluations, which remains pronounced. Instead, it captures how customer action becomes incorporated into the recovery process when participation occurs.

Taken together, we propose that CP mediates the relationship between service failure severity and recovery outcomes by serving as a behavioral pathway through which customers attempt to influence the trajectory of recovery.

Hypothesis 2. Customer participation (CP) mediates the effect of service failure severity on (a) satisfaction with the service encounter and (b) word-of-mouth.

2.3. The role of quality signals

The preceding section established that failure severity activates a discrepancy that motivates a behavioral response, and that CP

represents one pathway through which customers attempt to reduce this discrepancy. In this study, we focus on quality signals as a key contextual factor shaping how the relationship between service failure severity and CP unfolds. While the control-based framework explains why motivation to restore control arises, an important open question is how customers choose to restore that control. Specifically, contextual cues may shape whether customers intervene directly or rely on provider-driven resolution. Quality signals are examined as a critical boundary condition that may paradoxically discourage the participation that failure severity would otherwise elicit.

Hospitality firms frequently signal service quality through awards, certifications, and ratings (Sutherland et al., 2021). Drawing on signaling theory (Spence, 1973; Kirmani and Rao, 2000), such cues convey provider competence and reliability, thereby reducing uncertainty when evaluating service quality directly is difficult (Nicolau and Sellers, 2010). By projecting reliability and capability, quality signals shape customers' role expectations, including assumptions about how responsibility for managing failures should be allocated (Soberman, 2003).

Control theory suggests that while individuals are motivated to reduce discrepancies between desired and actual states (Carver and Scheier, 1981), it does not prescribe how control must be restored; rather, individuals select among alternative strategies for regaining control. In this context, quality signals do not diminish discrepancy-driven motivation. Instead, they shape how control is pursued. Visible quality cues communicate that the provider possesses both the capability and authority to resolve disruptions independently, reducing information asymmetry and shifting perceived responsibility toward the firm (Spence, 1973; Kirmani and Rao, 2000). When failure occurs in the presence of strong quality signals, customers may infer that recovery falls within the firm's domain of competence and rely more on provider-driven resolution rather than initiating direct intervention themselves (Hui and Toffoli, 2002).

Thus, while quality signals reduce perceived risk (Chen et al., 2024; Helm and Mark, 2007), they simultaneously raise the threshold at which failure severity triggers direct participatory action. Under high failure severity, the motivation to regain control remains strong. However, quality signals shape how customers interpret responsibility, shifting their chosen response toward relying on the provider to manage recovery rather than initiating direct intervention themselves, particularly when the need to regain control is less acute. In this way, quality signals do not eliminate participation but condition when and how it is enacted. Accordingly, we hypothesize:

Hypothesis 3. Quality signals moderate the effect of service failure severity on CP, such that the positive effect of failure severity on CP is weaker when quality signals are present.

2.4. The amplifying role of employee responsiveness

Employee responsiveness refers to the extent to which frontline employees acknowledge customer input and take visible action in response during service recovery. In high-contact hospitality contexts, responsiveness shapes how customers experience recovery interactions once participation has been initiated (Tung et al., 2017). Importantly, responsiveness is defined here as the acknowledgment and action taken in response to customers' input, rather than as politeness, courtesy, or generalized interaction quality.

When customers engage in CP, they attempt to influence the recovery process by articulating their needs, offering suggestions, or engaging with employees to resolve the issue. Whether these efforts translate into improved outcomes depends not only on customers' actions but also on how the firm responds. Responsive employees signal that customer input has been heard and acted upon (Tung et al., 2017), reinforcing the perception that their participation translates into improved recovery outcomes. Within the control-theoretic framework, experienced

responsiveness serves as a feedback mechanism through which customers assess whether their participation has reduced the discrepancy between the desired and actual service state. This feedback function enhances CP's capacity to restore a sense of control during recovery.

This conceptualization distinguishes responsiveness from related constructs such as interactional justice and double deviation (Bitner et al., 1990). Interactional justice emphasizes interpersonal treatment, respect, and dignity (e.g., Tax et al., 1998), whereas responsiveness captures whether customer input has an observable impact on the recovery process. Similarly, low responsiveness is not conceptualized as a second service failure or a failed recovery attempt, but rather as a feedback condition indicating whether participatory efforts are effectively integrated into recovery actions. When responsiveness is low, customers may perceive their participation as ineffective, weakening the positive effects of CP even if the substantive issue is ultimately resolved.

In line with this reasoning, employee responsiveness is modeled as primarily salient at the post-participation stage. While anticipatory expectations about responsiveness may inform participation decisions, customers at the initiation stage operate under uncertainty about how the firm will respond, relying instead on broader contextual cues such as quality signals to guide their action selection. Responsiveness becomes more diagnostic and consequential after CP has been initiated, when customers can directly evaluate whether their input has been acknowledged and acted upon. This stage-salience reflects the temporal structure of live service recovery encounters: firms rarely have a ready-made solution at the moment a failure is identified; rather, employees must assess the situation and determine feasible responses through interaction with the customer. Responsiveness becomes most diagnostic as employees engage with customer input, although anticipatory expectations about responsiveness may also shape participation decisions. We therefore hypothesize:

Hypothesis 4. Employee responsiveness moderates the effect of CP on (a) satisfaction and (b) word-of-mouth, such that the positive effects are stronger when responsiveness is high.

2.5. Conflicting signals: when quality cues undermine the impact of responsiveness on CP

Employee responsiveness can enhance the positive outcomes of CP by reinforcing the perceived impact and legitimacy of customer input. However, this effect may be limited by quality signals that shape expectations before recovery interactions begin. Together, these influences reflect a functional differentiation within the recovery process: quality signals operate as *ex ante* cues that shape how customers interpret the situation by influencing the perceived allocation of control between the customer and the firm, whereas employee responsiveness operates as *ex post* feedback, providing diagnostic information about whether enacted participation is acknowledged and meaningfully influences the recovery process. This distinction follows from control theory, which differentiates between cues that guide action selection and feedback that evaluates action effectiveness (Carver and Scheier, 1981). The differentiation is analytical rather than strictly temporal; both factors may be present throughout the encounter but exert their primary influence through distinct mechanisms.

As established above, quality signals shape whether customers perceive direct intervention as necessary by influencing the action-selection stage of the control process. By contrast, employee responsiveness serves a diagnostic function once participation has been initiated. When customers provide input, responsive behavior can validate their efforts and translate participation into improved recovery outcomes, serving the feedback-evaluation function within the control process. However, this feedback function becomes consequential only when participation occurs.

This asymmetry implies that quality signals primarily shape whether failure severity translates into participatory action, whereas

responsiveness shapes whether enacted participation translates into improved outcomes. When quality cues strongly signal provider competence, customers are less likely to initiate participation, thereby limiting the conditions under which responsiveness can exert its feedback-enhancing influence. In this sense, responsiveness is contingent on a prior decision shaped by quality signals, rather than being uniformly effective across all recovery situations. We therefore hypothesize:

Hypothesis 5. The indirect effects of service failure severity on (5a) satisfaction and (5b) word-of-mouth via CP are stronger when employee responsiveness is high, but this amplification is weakened in the presence of quality signals, which reduces the likelihood of initial participation.

3. Overview of studies

To test the proposed framework, we employ a multi-method research design combining qualitative, field, and experimental approaches

(Fig. 1). This design enables us to establish the core relationships, validate them across contexts, and examine the boundary conditions that shape when and how customer participation (CP) emerges and translates into recovery outcomes.

Studies 1–3 focus on establishing and validating the core process linking service failure severity, CP, and recovery outcomes. Study 1 uses the Critical Incident Technique to examine whether failure severity motivates CP (H1) and whether CP functions as a behavioral pathway linking severity to satisfaction and word-of-mouth (H2). Study 2 extends this test using large-scale online review data, providing behavioral validation of CP as an enacted response in real-world service settings. Study 3 offers causal evidence through a controlled experiment, replicating the mediating role of CP under tightly controlled conditions.

Building on this foundation, Study 4 introduces quality signals as a key boundary condition and examines their moderating role in shaping whether failure severity translates into CP (H3). This test establishes that participation is contingent not only on the level of disruption but also on contextual cues that shape perceived responsibility for recovery.

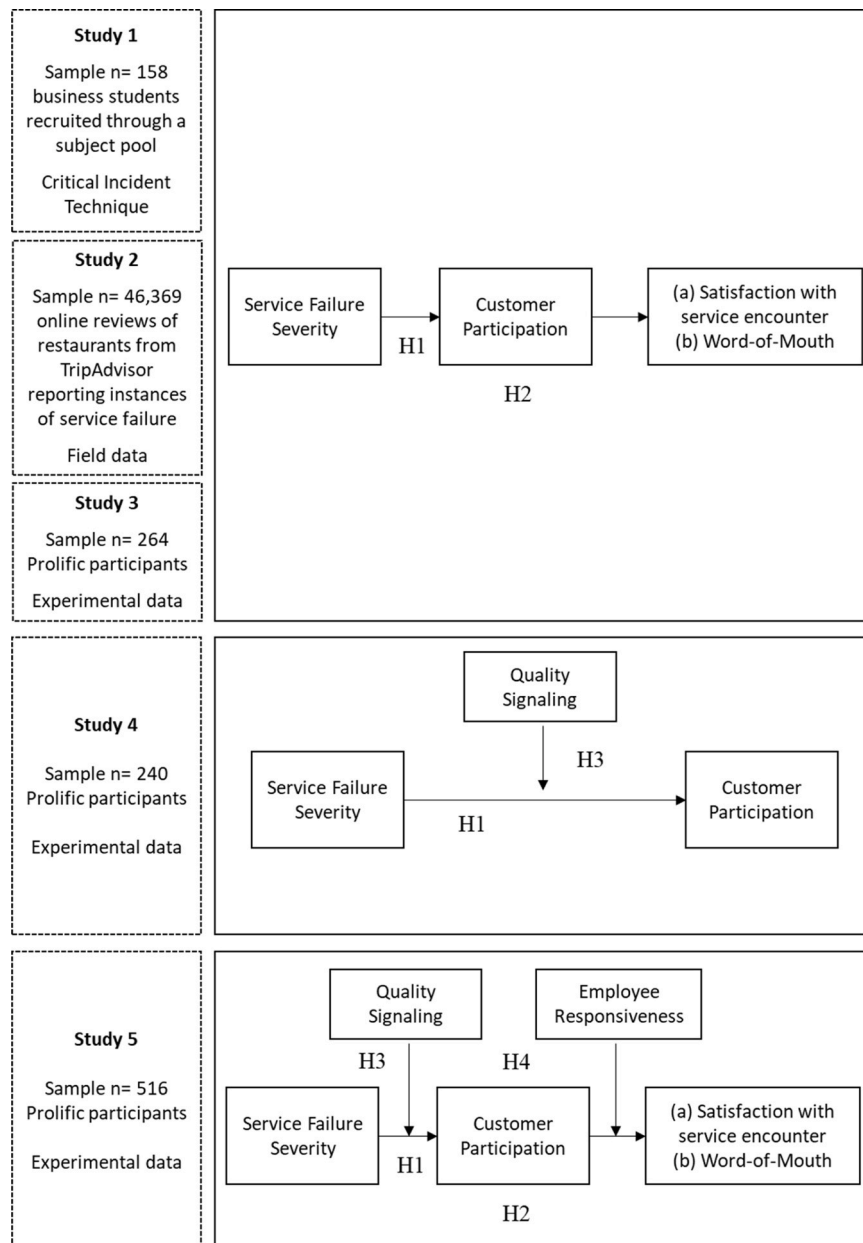


Fig. 1. Overview of empirical studies.

Finally, Study 5 provides a comprehensive test of the full model by jointly examining the roles of quality signals and employee responsiveness. This study tests whether responsiveness strengthens the effect of CP on recovery outcomes (H4) and whether this amplification is contingent on prior participation, which itself is conditioned by quality signals (H5). By integrating antecedents, mechanisms, and boundary conditions, the full set of studies provides a coherent test of CP as a situationally triggered behavioral pathway in service recovery.

4. Study 1: Establishing the mediating effect of CP

To deepen our understanding of CP in service recovery, our initial step involved employing the Critical Incident Technique (CIT) to gather data. Study 1 primarily aimed to empirically examine the relationship between service failure severity and CP (Hypothesis 1). At the same time, we aimed to investigate the role of CP as an underlying mechanism (Hypothesis 2) in the relationship between service failure severity and two specific service outcomes: (a) satisfaction with the service encounter, and (b) word-of-mouth.

4.1. Method

We recruited 167 participants from a subject pool managed by the business school at one of the largest Australian public universities. Each participant was asked to recall and describe in detail a recent service failure they experienced. Additionally, they were prompted to describe the actions taken by both themselves and the service provider during and after the incident to capture CP. Only narratives that provided a detailed account of the service failure and subsequent actions were retained for analysis, resulting in a final sample of 158 participants (63% females; $M_{age} = 19.95$, $SD_{age} = 2.08$). The narratives primarily featured service failures in retail outlets (28%), airline companies (18%), hotels (11%), and restaurants (9%).

Three independent assessors evaluated and coded the customer narratives to rate the severity of CP and service failure on a scale of 1–7, averaging the scores to compute variable values (see Supplement file A for scale items). Inter-rater reliability was assessed using an intra-class correlation coefficient (ICC) to evaluate the consistency of coders' ratings of the written statements. The resulting ICCs for service failure severity and CP, respectively ($ICC_{Severity} = 0.95$; $ICC_{CP} = 0.93$), were acceptable (Cicchetti, 1994), indicating a high degree of agreement. This suggests that CP and service failure severity were rated similarly by the independent assessors ($M_{Severity} = 4.22$, $SD_{Severity} = 1.39$; $M_{CP} = 3.46$, $SD_{CP} = 1.64$).

Participants rated their level of overall satisfaction with service encounter using three items adapted from Gelbrich (2011) and Maxham III, 2001 ($\alpha = 0.98$, $M = 2.92$, $SD = 1.70$) and word-of-mouth with 1 item from Maxham III, 2001 ('I would recommend this service provider to other people', $M = 3.46$, $SD = 1.82$). The relationship length with the service provider was used as a control variable and measured in months ($M = 34.27$, $SD = 50.6$).

4.2. Results

Hayes' PROCESS macro (Model 4) based on 10,000 bootstrap samples with a 95% bias-corrected interval was used to test the mediation effect of CP on the relationship between service failure severity and service outcomes (Hayes, 2022). Results show a significant positive effect of service failure severity on CP ($b = 0.37$; $t = 4.20$, $p < 0.001$), supporting Hypothesis 1. Results highlight a significant and positive effect of CP on (a) satisfaction with service encounter ($b = 0.33$, $t = 4.04$, $p < 0.001$) and (b) word-of-mouth ($b = 0.29$, $t = 3.43$, $p < 0.001$), while service failure severity has a significant negative effect on both service outcomes (satisfaction with service encounter: $b = -0.48$, $t = -5.08$, $p < 0.001$; word-of-mouth $b = -0.56$, $t = -5.71$, $p < 0.001$). Results also reveal a significant indirect effect of CP in the

relationship between service failure severity and (a) satisfaction with service encounter ($b = 0.12$, $CI = [0.05; 0.21]$), and (b) word-of-mouth ($b = 0.11$, $CI = [0.04; 0.20]$), supporting Hypothesis 2 for partial mediation as the direct effects of service failure severity on (a) satisfaction with the service encounter ($b = -0.48$; $t = -5.07$, $p < 0.001$) and (b) word-of-mouth ($b = -0.56$; $t = -5.71$, $p < 0.001$) are significant.

5. Study 2: Validating the mediating role of CP through behavioral data (Online reviews)

Study 2 aims to strengthen the validity of Hypothesis 2 by examining the mediating role of CP in service recovery, particularly in the relationship between service failure severity and service outcomes (satisfaction with the service encounter and word-of-mouth). To achieve this, we analyze field data sourced from online customer reviews that vividly portray instances of service failure. In this study, we conceptualize online reviews as enacted electronic word-of-mouth (eWOM) rather than as stated intentions to recommend. Unlike survey-based WOM measures, online reviews constitute a behavioral act through which customers publicly advocate for or against a service provider by sharing evaluative experiences with others (Hennig-Thurau et al., 2004; Litvin et al., 2008). The sentiment valence expressed in review text, therefore, captures the direction and intensity of advocacy embedded in this enacted WOM behavior. Notably, prior research demonstrates that review valence significantly influences consumer decision-making and purchase behavior (Chevalier and Mayzlin, 2006; You et al., 2015). Meta-analytic evidence shows that positive eWOM increases sales, while negative eWOM decreases them (Babić Rosario et al., 2016).

5.1. Method

We employed the web data crawler Apify to collect online restaurant reviews from TripAdvisor in Sydney, Australia. Our focus was on reviews posted in 2019 to eliminate potential biases introduced by the COVID-19 pandemic and associated lockdowns. In total, we scraped 49,902 reviews, including textual data and information on the restaurant's price levels (measured on a 1–4 scale: 1 = \$, 2 = \$\$, 3 = \$\$\$, 4 = \$\$\$\$), with no identifiers. We removed reviews written in languages other than English, resulting in a sample of 46,434 reviews. Following Pennebaker et al.'s (2015) recommendation to include only text samples with 10 or more words, we decided to analyze reviews with at least 15 words ($M = 72.64$, $SD = 68.42$) to ensure they captured the restaurant customer experience adequately. This criterion resulted in 46,369 usable reviews.

We analyzed these reviews using the Linguistic Inquiry and Word Count (LIWC) software to capture two constructs: service failure severity and CP (Pennebaker et al., 2015). To do this, we developed a custom dictionary tailored to our context. The dictionary development followed three steps. First, we created initial word lists for each construct by drawing on terms from academic literature and customer descriptions. This step produced dictionaries containing 12 words for severity and 14 for CP. Second, a research assistant and two academics validated the dictionaries for content. Based on their input, the final dictionaries contained 10 words each for severity and CP. The severity dictionary included terms indicative of strongly negative service experiences (e.g., disappoint*, unacceptable, ruined*). The CP dictionary included terms reflecting active, solution-oriented engagement (e.g., suggest*, ask*, offered). These dictionaries were then imported into LIWC for data analysis. In the third step, two researchers manually coded a random sample of 100 reviews. They rated the reviews for service failure severity and CP on a 1–5 scale. The interrater reliability ($ICC_{Severity} = 0.98$; $ICC_{CP} = 0.96$) was acceptable (Cicchetti, 1994). Inter-coder agreement helps to establish reliability by determining whether different judges consistently assign the same ratings (Lombard et al., 2004). There is a general agreement that an outcome above 0.90 is always acceptable (Nili et al.,

2020). For the dependent variables, the star rating (1–5) was used as a proxy for satisfaction with the service encounter.

To capture WOM valence, we performed a sentiment analysis (Hennig-Thurau et al., 2015) using the Valence Aware Dictionary and Sentiment Reasoner (VADER), a lexicon- and rule-based sentiment analysis tool tailored to social media contexts (Savelle et al., 2024). VADER combines dictionary and syntactical rules to assess the sentiment polarity and intensity of words, with pre-calculated sentiment scores for each word in its dictionary. This enables VADER to distinguish between positive and negative sentiment, particularly when handling idiomatic language commonly found in social media texts (Ribeiro et al., 2016). Compared to machine-learning models like Support Vector Machine (SVM), which require extensive training data, VADER offers better ease of use, interpretability, and accuracy (Borg and Boldt, 2020). VADER generates four scores for each sentence: negativity, neutrality, positivity, and a compound score. The compound score, ranging from −1 (extremely negative) to 1 (extremely positive), represents the overall sentiment of a sentence. These scores were averaged to assess the general sentiment of the reviews, serving as the dependent variable to capture word-of-mouth valence ($M = 0.45$, $SD = 0.30$). The validity of VADER-derived sentiment scores was assessed through comparison with human-coded sentiment ratings, as reported below. Restaurant price levels were included as a covariate to control for venue positioning. Supplement file B illustrates the sentiment analysis process.

To validate the automated text measures, two independent coders rated a random subsample of 100 reviews for severity, customer participation, and WOM sentiment. Inter-rater reliability was established using intraclass correlation coefficients (two-way mixed, absolute agreement, average measures): severity ICC = 0.98, CP ICC = 0.96, and WOM ICC = 0.95 (all $p < 0.001$). Averaged human ratings were then correlated with the corresponding automated scores. Correlations between human-coded and LIWC-derived scores were acceptable for severity ($r = 0.73$) and CP ($r = 0.70$). The correlation between human-coded sentiment and VADER compound scores was moderate ($r = 0.61$). These values are consistent with prior research applying automatic text analysis using rule-based and dictionary-based tools to review data (e.g., Aggarwal et al. 2025, reported $r = 0.59$ for VADER on physician reviews). The moderate correlations provide convergent evidence that the automated measures capture the intended constructs, while also suggesting that measurement noise from text-based proxies may contribute to the conservative effect sizes observed in this study (Funder and Ozer, 2019).

5.2. Results

To test the mediation role of CP on the relationship between service failure severity and (a) satisfaction with service encounter, (b) word-of-mouth, we used Hayes' PROCESS macro (Model 4) based on 10,000 bootstrap samples with a 95% bias-corrected interval (Hayes, 2022). Results confirm a significant positive association between service failure severity and CP ($b = 0.08$; $t = 12.98$, $p < 0.001$), consistent with Hypothesis 1. CP has a significant positive effect on (a) satisfaction with service encounter ($b = 0.01$; $t = 4.05$, $p < 0.001$) and (b) word-of-mouth ($b = 0.015$; $t = 18.69$, $p < 0.001$), whereas service failure severity has a significant negative effect on both service outcomes (satisfaction with service encounter: ($b = -0.30$; $t = -59.72$, $p < 0.001$); word-of-mouth: ($b = -0.06$; $t = -56.49$, $p < 0.001$). Most importantly, we find a significant indirect effect of CP in the relationship between service failure severity and (a) satisfaction with service encounter ($b = 0.0011$, CI = [0.0005; 0.0017]) and (b) word-of-mouth ($b = 0.001$, CI = [0.0009; 0.0014]). This supports Hypothesis 2 for partial mediation as the direct effects of service failure severity on (a) satisfaction with the service encounter ($b = -0.30$; $t = -57.89$, $p < 0.001$) and (b) word-of-mouth ($b = -0.06$; $t = -56.48$, $p < 0.001$) are significant. These findings conceptually validate the role of CP as an underlying solution-oriented behavioral response when measured via unobtrusive field data in

real-world settings, providing ecological validity.

6. Study 3: Validating the mediating role of CP through experimental data

The aim of Study 3 is to validate results from Studies 1 and 2 and provide further support for Hypotheses 1 and 2 by determining the mediating role of CP in service recovery in the relationship between service failure severity and service-related outcomes in a controlled experiment. In this and subsequent experiments, CP is operationalized as customers' willingness to engage in participatory recovery behaviors. This operationalization captures customers' propensity to intervene during an unfolding recovery episode, which precedes enacted participation when direct observation of behavior is constrained by experimental design.

6.1. Method

We recruited 270 UK consumers through the Prolific research panel. Six participants failed the attention check and were excluded from the subsequent data analysis, resulting in a final dataset of 264 respondents (45% males; $M_{age} = 46.96$, $SD = 13.01$). Participants were randomly assigned to a service failure severity condition (low severity: 15-minute delay; high severity: next available seating after a 2-hour delay) and were given a description of a service episode involving a mishap with a restaurant booking (see Appendix A for scenarios).

After reading the scenario, participants rated their likelihood of participating in the service recovery process (five items adapted from Chan et al., 2010; $\alpha = 0.93$, $M = 3.84$, $SD = 1.83$), their level of satisfaction with service encounter (four items adapted from (Maxham III, 2001; Gelbrich, 2011); $\alpha = 0.96$, $M = 2.96$, $SD = 1.63$) and word-of-mouth (three items adapted from (Maxham III, 2001; Gelbrich, 2011); $\alpha = 0.97$, $M = 3.19$, $SD = 1.60$) (see Supplement File C). Participants also rated the scenario as realistic and believable, indicating that they were able to engage meaningfully with the service episode ($M = 5.77$, $SD = 1.03$). All items were measured on a 7-point scale and averaged to compute variable scores.

To assess the manipulation of service failure severity, we conducted a one-way ANOVA. Results showed a substantial difference between low severity ($M_{Severity, Low} = 2.42$, $SD = 1.29$) and high severity conditions ($M_{Severity, High} = 6.35$, $SD = 0.83$), $F(1, 262) = 864.39$, $p < 0.001$, $\eta_p^2 = 0.77$, confirming the effectiveness of the manipulation.

6.2. Results

We conducted a mediation analysis using Hayes' PROCESS macro (Model 4) based on 10,000 bootstrap samples and 95% bias-corrected intervals. Results showed that severity of service failure had a positive relationship with CP ($b = 2.26$, $SE = 0.17$, 95% CI [1.91, 2.61]), while CP was positively associated with satisfaction ($b = 0.10$, $SE = 0.04$, 95% CI [0.01, 0.18]) and WOM ($b = 0.12$, $SE = 0.04$, 95% CI [0.03, 0.21]). The indirect effect of CP on the relationship between severity of service failure and satisfaction was significant ($b = 0.22$, 95% CI [0.04, 0.39]), as was the indirect effect between severity and WOM ($b = 0.28$, 95% CI [0.11, 0.45]). The direct effects of service failure severity on satisfaction ($b = -2.77$, 95% CI [-3.08, -2.46]) and word-of-mouth ($b = -2.67$, 95% CI [-3.00, -2.34]) remained strongly negative and significant, confirming partial mediation. Consistent with the theoretical framing developed in H2, the dominant force remains with the negative impact of failure severity on outcomes, while CP operates as a contingent behavioral pathway that partially offsets this effect when participation occurs.

Results from the controlled experiment conducted in Study 3 confirm and extend the results obtained from Studies 1 and 2. The results demonstrate that CP serves as a solution-oriented behavioral response that functions as a pathway through which service failure severity

influences outcomes. The strong negative direct effects of severity on both outcomes confirm that CP does not reverse the fundamental impact of service failure, but operates as a contingent pathway whose contribution depends on the conditions examined in subsequent studies.

7. Study 4: Establishing the moderating effect of quality signaling

Study 4 aims to examine the moderating effect of quality signaling on the relationship between service failure severity and CP, as proposed in Hypothesis 3. By focusing on this aspect, we aim to elucidate the conditions under which the relationship between service failure severity and CP may vary.

7.1. Method

We recruited 250 UK participants through Prolific to participate in an online scenario-based survey (final $n = 240$ [10 failed attention checks], 42% males; $M_{age} = 45.88$, $SD_{age} = 13.68$). Respondents were randomly allocated to conditions in a 2 (No quality signaling, Quality signaling) $\times$ 2 (Low service failure severity, High service failure severity) between-subjects design. First, participants were presented with an image of a restaurant. We used quality awards as signaling stimuli based on Nicolau and Sellers (2010). In the quality signaling condition, the image displayed a screen with a TripAdvisor certificate of quality and a message stating that past customers had rated the restaurant's service as excellent (see Appendix A for scenarios). TripAdvisor is a widely recognized quality indicator in the hospitality industry, and its certification display reflects realistic marketplace practices (Jeacle and Carter, 2011). Participants were then asked to imagine that they had booked a dinner at that restaurant. However, the dinner was delayed by 15 min (low service failure severity) or delayed by 2 h and moved to the next available seating (high service failure severity). Participants rated the extent to which they would have engaged with the restaurant employee to participate in the service recovery process (five items adapted from Chan et al., 2010; $\alpha = 0.85$, $M = 3.99$, $SD = 1.31$). Items were measured on a 7-point scale and averaged to compute variable scores (see Supplement File C for scale items).

We adopted three items from Hess et al., (2003) as the manipulation check for service failure severity ($\alpha = 0.98$, $M = 4.34$, $SD = 2.19$). We also measured the perceived presence of quality signals using four items created based on the conceptualization of quality signals (Nicolau and Sellers, 2010; $\alpha = 0.96$, $M = 5.15$, $SD = 1.36$; sample items: "TripAdvisor acknowledged the excellence of this restaurant", "This restaurant is rated highly by past customers"). Manipulation checks for service failure severity were performed through a 2 (Low service failure severity, High service failure severity) $\times$ 2 (No quality signaling, Quality signaling) ANOVA. This analysis reveals only a main effect of severity ($M_{Severity_Low} = 2.52$, $SD = 1.37$; $M_{Severity_High} = 6.11$, $SD = 1.14$; $F(1, 238) = 487.66$, $p < 0.001$, $\eta_p^2 = 0.67$). The same 2 $\times$ 2 ANOVA was conducted for the quality signaling manipulation check, which revealed only a main effect of quality signaling ($F(1, 238) = 338.78$, $p < 0.001$, $\eta_p^2 = 0.59$). Participants in the 'no quality signaling' (vs. 'quality signaling') condition reported perceiving lower levels of quality signaling ($M_{NoQS} = 4.13$, $SD = 0.99$; $M_{QS} = 6.22$, $SD = 0.75$). Participants rated the scenario as highly realistic and believable, suggesting they could engage meaningfully with the service episode ($M = 5.82$, $SD = 0.99$).

7.2. Results

We conducted a moderation analysis to assess how service failure severity affects CP in service recovery under different quality signaling conditions. We tested the interaction effect using the PROCESS Model 1 (Hayes, 2022) with 10,000 bootstrap samples. Results confirm a significant interaction between service failure severity and quality signaling ($b = -1.44$; $t = -4.93$, $p < 0.001$), lending support to

Hypothesis 3 and confirming the moderating effect of quality signaling in the relationship between service failure severity and CP. Specifically, in the no quality signaling condition, we record a positive and significant effect of service failure severity on CP ($b = 1.82$; $t = 8.93$, $p < 0.001$; $CI = [1.42, 2.22]$), while in the quality signaling condition, there is no significant effect of service failure severity on CP ($b = 0.38$; $t = 1.81$, $p = 0.07$; $CI = [-0.03, 0.79]$) (Fig. 2).

8. Study 5: Assessing the role of quality signaling and employee responsiveness in CP

Study 5 encompasses two primary objectives. First, it aims to establish causal evidence for the interactive effect of employee responsiveness on the relationship between CP and service outcomes. Second, it seeks to examine the extent of the indirect effect of service failure severity on service outcomes via CP within the context of service recovery, while considering two boundary conditions: quality signaling and employee responsiveness.

8.1. Method

A total of 530 UK participants ($n = 516$ after removing 14 who failed attention checks; $M_{age} = 43.60$ years, $SD = 13.2$, 49.6% females) were recruited from the online panel Prolific. In this study, participants were first randomly assigned to one of the four experimental conditions based on a 2 (Low service failure severity vs. High service failure severity) $\times$ 2 (No quality signaling vs. Quality signaling) between-subjects design (see Appendix A for scenarios). Participants were instructed to imagine that they had booked a tropical resort holiday and paid an additional 20% for upgraded amenities and complimentary services. After reading this initial information, participants completed a short distraction activity and attention checks to ensure engagement with the scenario and comprehension of the upcoming task.

Following this, participants proceeded to the role-play portion of the scenario, simulating the period leading up to their arrival at the resort. At this stage, the participants received a courtesy text message that introduced the severity of service failure manipulation. In the high-severity condition, the message confirmed their booking without including any of the upgrades they had paid for. In the low-severity condition, the message confirmed the booking along with only a portion of the upgrades.

After being exposed to the service failure stimuli, the scenario directly places participants arriving at the resort reception, where they were exposed to the quality signaling manipulation. In the quality signaling condition, the reception image displayed a TripAdvisor certificate indicating that previous guests had rated the resort as excellent. In the no-signal condition, the same screen displayed a neutral "Welcome" message. Participants then responded to the CP measure used in previous studies ($\alpha = 0.85$, $M = 5.72$, $SD = 1.04$) and described briefly how they would approach resolving the issue.

After completing this section, participants were then exposed to the responsiveness manipulation. Participants were informed that the receptionist acknowledged their feedback and either resolved the booking issue immediately (high responsiveness) or indicated that the matter required managerial approval the next day (low responsiveness).

They then rated their satisfaction with the service encounter ($\alpha = 0.95$, $M = 3.54$, $SD = 1.84$) and their likelihood to recommend the hotel ($\alpha = 0.98$, $M = 3.73$, $SD = 1.83$) (see Supplement File C for scale items). Perceptions of service failure severity were measured using three items adapted from Hess et al., (2003) ($\alpha = 0.89$, $M = 5.38$, $SD = 1.35$). Perceptions of quality signaling were measured using four items adapted from Nicolau and Sellers (2010) ($\alpha = 0.94$, $M = 5.35$, $SD = 1.11$). Employee responsiveness was assessed using two items ("Would you consider your problem to have been solved (i) with delay/promptly; (ii) unacceptably/acceptably"; $r = 0.79$, $p < 0.001$, $M = 4.34$, $SD = 2.42$). Participants rated the scenario as realistic and believable ($M = 6.22$, SD

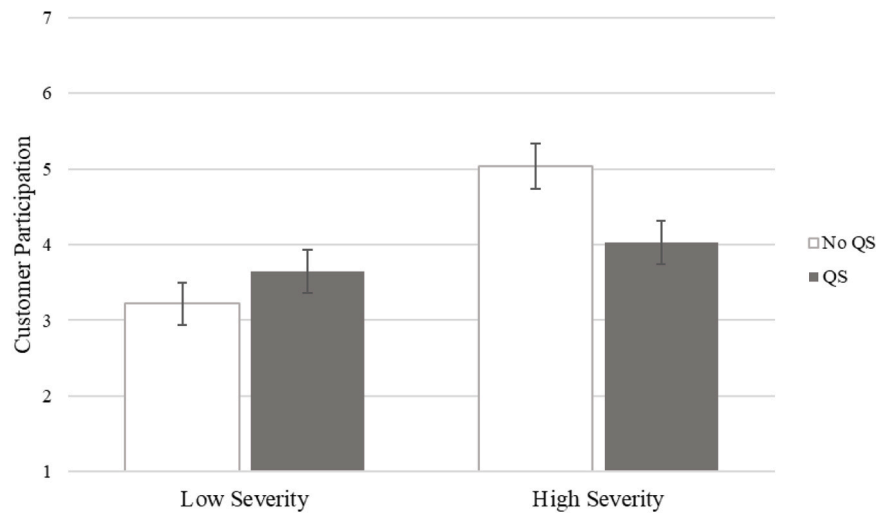


Fig. 2. Interaction effect between service failure severity and quality signaling on CP (Study 4).

= 0.99).

The sequential structure of Study 5 is designed to capture the functional differentiation within the control-based process, distinguishing between factors that shape participation initiation and those that influence how participation translates into recovery outcomes. By measuring CP prior to the responsiveness manipulation, the design ensures that the initiation of participation is not contaminated by knowledge of how the firm will respond, allowing a clean test of whether responsiveness shapes the translation of participation into outcomes (its theorized *ex post* feedback-evaluative role on CP) rather than the decision to participate itself. This sequencing follows established principles for moderated mediation designs in which the mediator must be assessed prior to the second-stage moderator (Hayes, 2022) and reflects the temporal structure of live service recovery encounters, where responsiveness becomes most diagnostic as the employee engages with the customer's input, rather than at the point of failure.

Consistent with recommendations for conducting manipulation checks in factorial designs (Perdue and Summers, 1986), a 2×2 ANOVA was performed on the severity manipulation check. The analysis revealed a significant main effect of severity ($M_{\text{Low}} = 4.87$, $SD = 1.33$, $n = 259$; $M_{\text{High}} = 5.88$, $SD = 1.16$, $n = 257$; $F(1, 512) = 82.68$, $p < 0.001$, $\eta_p^2 = 0.14$). Neither the main effect of signaling nor the interaction term was significant, indicating that the severity manipulation operated as intended and independently of the other factor. A corresponding 2×2 ANOVA on perceived quality signaling showed a significant main effect of the signal manipulation ($F(1, 512) = 431.69$, $p < 0.001$, $\eta_p^2 = 0.43$), with participants in the no-signal condition reporting lower perceived quality signaling ($M = 4.60$, $SD = 0.87$) than those in the signal-present condition ($M = 6.10$, $SD = 0.77$). Again, neither the severity main effect nor the interaction effect was significant, confirming the independence of the signaling manipulation. Finally, the manipulation check for employee responsiveness indicated a substantial difference between low responsiveness ($M = 2.12$, $SD = 1.19$) and high responsiveness ($M = 6.56$, $SD = 0.69$), $F(1, 514) = 2655.30$, $p < 0.001$, confirming that this manipulation was also successful.

8.2. Results

Moderated mediation analyses were conducted using PROCESS Model 21 with 10,000 bootstrap samples (Hayes, 2022) to test Hypotheses 4 and 5. When satisfaction was entered as the dependent variable, the overall model revealed a significant index of moderated moderated-mediation ($IMM = -0.12$, $SE = 0.06$, 95% CI $[-0.26, -0.01]$), that was validated by significant interaction effects: service failure severity $\times$ quality signaling ($b = -0.40$; $p < 0.05$) and CP

$\times$ employee responsiveness ($b = 0.29$; $p < 0.001$). Consistent with Study 4, the effect of service failure severity on CP was positive and significant when no quality signal was present ($b = 0.52$, $SE = 0.12$, 95% CI $[0.28, 0.78]$) but was not significant when a quality signal was displayed ($b = 0.13$, $SE = 0.13$, 95% CI $[-0.12, 0.37]$).

Furthermore, the index of conditional moderated mediation is statistically significant when employee responsiveness is high ($IMM = -0.07$, $SE = 0.04$, CI $[-0.17, -0.003]$). Specifically, high employee responsiveness strengthens the positive effect of CP on satisfaction with service encounter ($b = 0.17$, $SE = 0.07$, CI $[0.04, 0.31]$), whereas low employee responsiveness results in a non-significant effect ($b = -0.11$, $SE = 0.06$, CI $[-0.20, 0.10]$), providing support for Hypothesis 4a. In high responsiveness condition, the indirect effect of CP between service failure severity and satisfaction is positive and significant in the absence of a quality signal in the service environment ($b = 0.10$, $SE = 0.04$, CI $[0.01, 0.18]$), but not statistically significant when a quality signal is present ($b = 0.022$, $SE = 0.03$, CI $[-0.02, 0.097]$), thus providing partial support to Hypothesis 5a.

A parallel model using word-of-mouth intentions yielded a similar pattern. The overall index of moderated moderated-mediation was significant ($IMM = -0.13$, $SE = 0.07$, 95% CI $[-0.29, -0.001]$). The interaction between CP and responsiveness remained significant ($b = 0.32$, $p < 0.001$). CP predicted higher word-of-mouth intentions when responsiveness was high ($b = 0.22$, $SE = 0.07$, 95% CI $[0.08, 0.36]$), but not when responsiveness was low ($b = -0.10$, $SE = 0.07$, 95% CI $[-0.24, 0.03]$), supporting Hypothesis 4b. The conditional indirect effect of severity on word-of-mouth through CP was significant only when no quality signal was present, and responsiveness was high ($b = 0.12$, $SE = 0.05$, 95% CI $[0.01, 0.18]$), providing partial support for Hypothesis 5b.

9. General discussion

This research reframes customer participation (CP) as a situationally triggered, control-restoration response that emerges during the recovery process. Across five studies, we demonstrate that severe service disruptions, rather than firm invitations, prompt customers to intervene when recovery stalls and customers' sense of control is threatened. CP operates as a solution-oriented behavioral response through which customers attempt to regain influence over stalled recovery processes. This process view shifts attention from post-hoc evaluation to customer behavior during recovery, explaining when customers are more or less likely to engage in constructive participation.

Participation is neither automatic nor uniformly beneficial. Quality signals create a conditional participation paradox: cues designed to convey competence (such as awards and certifications) attenuate the

extent to which service failure severity translates into customer participation by shifting responsibility toward the firm. This does not eliminate participation but reduces constructive intervention during recovery, while remaining distinct from other responses such as complaints, exit, or withdrawal.

Employee responsiveness plays a stage-salient role by enhancing the extent to which CP translates into improved recovery outcomes once enacted, through signaling acknowledgment and action on customer input. However, responsiveness cannot compensate when quality signals have shifted customers away from direct participatory action. This asymmetry reveals that quality cues shape CP initiation while responsiveness shapes the extent to which CP translates into improved recovery outcomes.

9.1. Theoretical contributions

This research makes four theoretical contributions to the service recovery and customer participation (CP) literature.

First, we reconceptualize CP in service recovery as a situationally triggered, solution-oriented behavioral response rather than as a proactive form of value co-creation or a function of stable customer traits. Prior research has largely treated CP as an invited or role-framed activity that enhances outcomes when customers are willing and prepared to participate (Dong et al., 2008; Vargo and Lusch, 2016). In contrast, we show that CP emerges during recovery when service failures threaten customers' sense of control and recovery progress appears uncertain. Drawing on control theory (Carver and Scheier, 1981; Hui and Bateson, 1991), we conceptualize CP as a solution-oriented behavioral response through which customers seek to restore influence over the unfolding recovery process. This reframing shifts the attention of participation as a firm-invited system feature to participation as a situationally triggered, solution-oriented response enacted during service recovery.

Second, this research identifies an explanatory gap in dominant recovery frameworks that CP addresses. Expectancy-disconfirmation (Oliver, 1980) and justice theory (e.g., Tax et al., 1998) explain how customers evaluate recovery outcomes, but they do not account for variation in customer action during recovery itself, specifically, why some customers intervene constructively while others remain passive. By positioning CP within a control-theoretic process, our framework explains this variation: failure severity activates a motivation to restore control, but whether that motivation produces participatory action depends on contextual cues such as quality signals. This positions CP as complementary to evaluative frameworks, addressing a distinct stage of the recovery process that existing theories leave unspecified.

Third, we identify a conditional participation paradox associated with quality signals. Whereas signaling theory emphasizes the uncertainty-reducing and confidence-enhancing effects of quality cues (Spence, 1973; Kirmani and Rao, 2000), we show that in recovery contexts, strong quality signals can attenuate the likelihood that service failure severity triggers CP. By shaping role expectations and shifting perceived responsibility for recovery towards the provider, quality signals reduce customers' marginal propensity to intervene constructively. This extends signaling theory by demonstrating that signals influence not only beliefs and evaluations, but also behavioral thresholds for participation during service breakdowns. The findings are in line with control theory to show quality signals can influence the control strategy a customer undertakes.

Fourth, we contribute a stage salient temporal account of contextual influences in service recovery by distinguishing the roles of quality signals and employee responsiveness. We show that quality signals shape whether failure severity translates into participatory action, whereas employee responsiveness serves a diagnostic function once CP is enacted, conditioning the extent to which CP translates into improved outcomes. This sequencing reveals a critical boundary condition for responsiveness: its benefits are contingent on participation having already occurred. By unpacking this asymmetry, we advance a more

dynamic understanding of how contextual cues jointly shape recovery processes and outcomes. Collectively, these contributions position CP as a situationally triggered, solution-oriented behavioral response embedded within recovery episodes, moving beyond static evaluative views to explain real-time customer action during high-stakes service failures.

9.2. Managerial implications

This research offers actionable guidance for hospitality firms seeking to design recovery systems that are not only operationally effective but also psychologically aligned with customer behavior during severe service failures. Across our studies, customer participation (CP) emerges as a situational response that can improve recovery outcomes; yet whether customers participate and whether participation helps depends on cues embedded in the recovery environment.

First, firms should recognize that customers facing high-severity failures may be willing to participate in recovery, particularly when progress appears stalled, and the experience matters. However, this willingness depends on whether the context signals that participation is welcome and appropriate. Hospitality firms can facilitate constructive CP by creating visible, low-effort avenues for input at key touchpoints (e.g., in-room tablets, mobile prompts, or table cards) that explicitly invite guidance, such as: "Help us make it right: tell us what matters most now."

Second, firms should be aware that quality signals can attenuate customers' likelihood of initiating participation by shifting perceived responsibility for recovery toward the provider. While certifications, awards, and rankings build confidence, they may also lead customers to assume that the firm will handle recovery autonomously. To counter this effect, firms should balance reputational cues with participatory messaging that frames customer input as part of maintaining service excellence rather than as a challenge to it. For example, a hotel might highlight its accolades in pre-arrival communication while signaling openness during service: "We are proud of our awards, and your feedback helps us live up to them."

Third, employee responsiveness remains critical. Although responsiveness cannot compensate when participation does not occur, it can meaningfully amplify the benefits of CP once customers choose to engage. Training should therefore emphasize acknowledgment and visible action on customer input, not just procedural resolution. When customers see that their contributions shape recovery actions, CP is more likely to restore a sense of influence, which in turn translates into higher satisfaction and positive word-of-mouth.

Taken together, recovery should be designed as a shared process. Firms that combine quality signaling with explicit participatory cues and responsive frontline employee behavior are better positioned to harness CP when it emerges and improve recovery outcomes following severe service failures.

9.3. Limitations and future research

Several limitations suggest promising research directions. First, while using real-world cues like TripAdvisor certifications, future work could test fictional, platform-specific, or digital signals to isolate effects across service contexts. Second, we establish CP as a solution-oriented behavioral response that functions as a mediator linking severity to outcomes, but we did not measure psychological mediators, such as disconfirmation or justice perceptions. These findings position CP alongside, rather than replacing, these evaluative mechanisms. Future studies should jointly model CP with fairness, disconfirmation, emotion, or trust to examine how behavioral intervention interacts with post-hoc evaluation. Third, we examined quality signals and responsiveness as key boundary conditions. Additional factors, such as prior brand experience, relationship tenure, and customer expertise, are likely to influence intervention decisions and how participation is received, thereby

enhancing ecological realism. Fourth, while our framework models employee responsiveness as the diagnostic feedback condition following participation. Future research could examine whether expectations of employee responsiveness shape the initiation of CP. Additionally, while the sequential design in Study 5 isolates the theoretically focal mechanisms by separating quality signal exposure, participation, and responsiveness into distinct stages, naturalistic recovery encounters could be more fluid, with these cues potentially co-occurring and interacting dynamically. Future research using field-based or longitudinal designs could examine how quality signals and responsiveness jointly evolve across the recovery episode. Finally, while experiments establish causality, longitudinal field research could track CP evolution across repeated failures or extended episodes, clarifying how responsiveness shapes the extent to which participation translates into outcomes post-enactment and customers' willingness to sustain participation over time.

CRediT authorship contribution statement

Wei Wei Cheryl Leo: Writing – original draft, Conceptualization.
Isabella Maggioni: Writing – review & editing, Investigation, Formal

analysis, Conceptualization. **Yelena Tsarenko:** Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Agung Yoga Sembada:** Writing – review & editing, Investigation, Formal analysis.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the author(s) used ChatGPT to support the refinement of text, including grammar, phrasing, and clarity. The author(s) critically reviewed and edited all AI-generated content and take(s) full responsibility for the integrity and accuracy of the final manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Scenarios

Study 3

Low service failure severity

On a Friday night, you booked dinner at a restaurant. You made a booking at 7:00 pm for you and your partner.

You arrive at the restaurant at 7:00 pm sharp and the staff member at the counter welcomes you. After checking your details, the staff member explains that there was a small mix-up in how the bookings were sequenced and your table is not yet ready. Your reservation is still confirmed, but your table will only be available at around 7:15 pm.

High service failure severity

On a Friday night, you booked dinner at a restaurant. You made a booking at 7:00 pm for you and your partner.

You arrive at the restaurant at 7:00 pm sharp and the staff member at the counter welcomes you. After checking your details, the staff member explains that there has been an error in the booking system, affecting your reservation. Your reservation is still confirmed, but due to this scheduling breakdown your table will only be available for the next seating, at around 9:00 pm.

Study 4

Quality signaling conditions

Quality signaling



No quality signaling



Low service failure severity

On a Friday night, you booked dinner at the restaurant you have just seen in the picture. You made a booking at 7:00 pm for you and your partner.

You arrive at the restaurant at 7:00 pm sharp and the staff member at the counter welcomes you. After checking your details, the staff member explains that there was a small mix-up in how the bookings were sequenced and your table is not yet ready. Your reservation is still confirmed, but your table will only be available at around 7:15 pm.

High service failure severity

On a Friday night, you booked dinner at the restaurant you have just seen in the picture. You made a booking at 7:00 pm for you and your partner.

You arrive at the restaurant at 7:00 pm sharp and the staff member at the counter welcomes you. After checking your details, the staff member explains that there has been an error in the booking system, affecting your reservation. Your reservation is still confirmed, but due to this scheduling

breakdown your table will only be available for the next seating, at around 9:00 pm.

Study 5

For your vacation, you decide to book a trip to a tropical resort in the Pacific Islands. You book an accommodation close to the beach and the main shopping strip. You have opted for the Value Package deal, which was a three-night stay, and costs \$150 per night.

A few days before your trip, you received an email from the hotel:

Dear [Name],

Would you like to add a little luxury to your upcoming trip?

For only \$100, you can upgrade your entire booking to the Premium Package (valued at \$200) and add that special something to your stay with us. You will enjoy an extra level of comfort in our larger Bungalow room, with a private balcony overlooking the beach. You will also receive: Welcome drinks on your arrival, One complimentary dinner at the hotel restaurant, Extended check-out time at the end of your stay.

Low service failure severity

You are excited to go on your upgraded trip! The day arrived and you landed safely at your destination. In the taxi, you received a text message from the hotel.

Dear [Name],

Thank you for staying with us, your room is now ready for check-in. Your benefits await your stay: Bungalow room, Welcome drinks at your arrival, One complimentary dinner at the hotel restaurant.

You noticed there seems to be a mistake here, but you cannot use your phone to call them back since you are overseas.

High service failure severity

You are excited to go on your upgraded trip! The day arrived and you landed safely at your destination. In the taxi, you received a text message from the hotel.

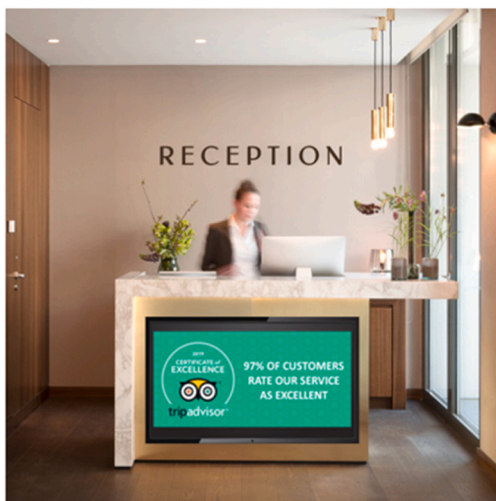
Dear [Name],

Thank you for staying with us, your Value Package room selection is now ready for check-in.

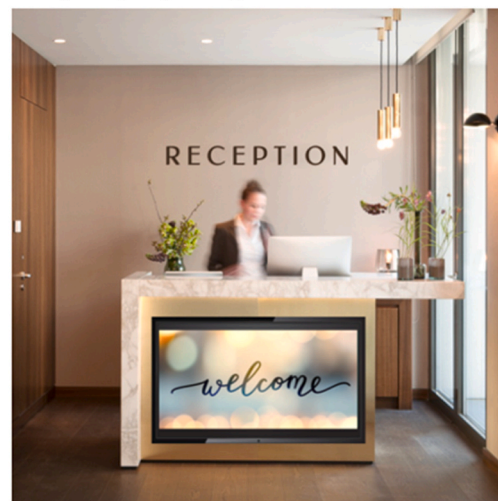
You noticed there seems to be a mistake here, but you cannot use your phone to call them back since you are overseas.

Quality signaling conditions

Quality signaling



No quality signaling



You have now arrived at the hotel, where a receptionist is assisting you with the check-in process. You mention the upgraded booking you paid for and show her the mistake in the text message you received.

Appendix B. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.ijhm.2026.104809](https://doi.org/10.1016/j.ijhm.2026.104809).

Data availability

The data that has been used is confidential.

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